

SYSTEMS & PROGRAMMING

- Python
- SQL
- Distributed Systems
- API Services
- Stream / Batch
- Performance Testing

INFERENCE & LLM SYSTEMS

- vLLM
- OpenAI-Compatible APIs
- LoRA / QLoRA
- PEFT
- Transformers
- 4-bit / 8-bit Quantization
- Distributed Training

CLOUD & ORCHESTRATION

- GCP
- GKE
- Vertex AI
- Kubernetes
- Docker
- Helm
- Istio
- Cloud Build
- ArgoCD
- Google KCC

RELIABILITY & DELIVERY

- CI/CD
- GitOps
- HPA Autoscaling
- Prometheus
- Grafana
- Blue-Green Delivery
- Load Testing
- SLA Operations

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024
Generative AI Certified Professional
- Microsoft Certified
Azure AI Engineer Associate
- Microsoft Certified
Azure Data Scientist Associate
- Google Cloud
Certified Generative AI Leader

EDUCATION

- B.E. Electrical & Electronics
Andhra University | 2012 - 2016

ML METHODS

- Scikit-learn
- PyTorch
- TensorFlow
- Keras
- DenseNet
- Classification
- Predictive Modelling
- Unsupervised Learning
- Feature Engineering
- Transfer Learning
- Anomaly Detection
- NLP
- Time-Series Forecasting

Production AI and inference engineer with 10+ years building reliable software and high-throughput ML systems. At Best Buy, built a shared LoRA/QLoRA training, evaluation, artifact-lifecycle, and vLLM serving platform on GKE that replaced external LLM APIs and supports estimated annual API-cost savings of **~\$300K**. The framework uses 4-bit and 8-bit quantization; used QLoRA for parameter-efficient fine-tuning. Separately operates real-time ML infrastructure with feature pipelines handling **40,000 events/sec** at peak and a **200 ms scoring SLA**. Earlier built continuously retrained billion-row GPU pipelines on an 8xA100 DGX. Hands-on with Python, vLLM, GPU ML systems, Kubernetes, safe rollout, observability, performance diagnosis, and end-to-end production ownership.

EXPERIENCE

Lead Artificial Intelligence Engineer

Best Buy India

Jul 2025 - Present | Bengaluru

- Architected a shared production platform for config-driven LoRA/QLoRA training, automated post-training evaluation, artifact lifecycle, and vLLM OpenAI-compatible model serving on GKE.
- Enabled fine-tuned adapters to hot-load into a shared vLLM base model rather than require a separate deployment per model; self-hosted services support estimated annual API-cost savings of **~\$300K**. Kept model inputs company-managed and reduced external API exposure during peak demand.
- Built the fine-tuning framework with 4-bit and 8-bit quantization; used QLoRA for experiments on shared GPU capacity.
- Designed autoscaled delivery with HPA, Istio traffic routing, blue-green releases, Cloud Build, ArgoCD, Google KCC, Prometheus, and Grafana; owned load testing and operational diagnosis.
- Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.

Lead ML Engineer — Machine Learning & MLOps

Wipro

Aug 2023 - Jul 2025 | Hyderabad

- Built distributed training and inference pipelines on NVIDIA DGX (8xA100), processing **billion-row** AWS CloudTrail datasets with Morpheus, RAPIDS, MLflow, continuous retraining, and Grafana alerting.
- Engineered phishing classification at **2,400 emails/sec** - 2x prior throughput - and multiformat abuse-inbox processing integrated with Microsoft Sentinel and production security workflows.
- Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery; completed **300+ résumé screens** and conducted **50+ technical interviews**.

Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 - Sep 2023 | India

- Led end-to-end Azure ML product delivery for BamaGruppen, increasing inspections from **130 to 160 trucks/day**, improving low-quality detection 17%, and generating **\$2.5M annual savings**.
- Built production services and automation with Python, Azure Functions, Logic Apps, Azure DevOps, CI/CD, and Google OR-Tools; translated ambiguous client needs into operated systems.

PROJECTS

Shared LLM Training & Serving

Best Buy | GKE | vLLM | LoRA

A shared base-model approach for fine-tuning adapters, automated evaluation, artifact management, autoscaled inference, routing, safe rollout, and full-stack observability.

Real-Time Model Serving

Best Buy | Seldon | 40k feature events/sec

Streaming feature engineering and low-latency inference with Kubernetes autoscaling, secure endpoints, load testing, telemetry, GitOps delivery, and a 200ms SLA.

GPU STACK

- CUDA GPU Libraries
- RAPIDS cuDF / CuPy
- Multi-GPU Training

OPTIMISATION & SIMULATION

- Google OR-Tools
- Vehicle Routing
- Mixed-Integer Programming
- Constraint Modelling
- Operations Research
- Assignment Models
- Supply-Chain Optimization
- Scenario Analysis

CLOUD & MANAGED ML

- GCS
- Artifact Registry
- Azure ML
- Azure DevOps
- Function Apps
- Logic Apps

DATA & ML LIFECYCLE

- Apache Beam
- Dataflow
- Pub/Sub
- Vertex AI Feature Store
- Streaming Pipelines
- Training Pipelines
- Automated Evaluation
- Experiment Tracking
- Model Registry
- Model Versioning
- Continuous Training

LANGUAGES & DATA

- Pandas
- NumPy
- MS SQL Server
- IBM DB2
- Power BI

ENGINEERING LEADERSHIP

- Architecture Reviews
- Technical Mentoring
- Technical Hiring
- Team Building
- Stakeholder Alignment
- Cross-Functional Delivery

ACHIEVEMENTS

15 honors and awards across Best Buy, Wipro, Accenture, and Infosys.
Best Buy Talent Quotient (2026): cybersecurity and fraud delivery.
Best Buy client AML Hall of Fame (2024): DGX delivery during the Wipro engagement.

ADDITIONAL PROJECTS & EXPERIENCE

GPU ML Pipelines

Wipro | DGX | Billion-row scale

Continuously retrained behavioral models and high-throughput classification pipelines with GPU acceleration, lifecycle tracking, monitoring, and analyst-facing integrations.

Inference Internals

Personal project | PyTorch | In progress

Building KV-cache, paged-attention, and speculative-decoding concepts from scratch alongside a config-driven LoRA-to-evaluation-to-vLLM serving pipeline.

EARLIER EXPERIENCE & ACHIEVEMENTS

Data Scientist

GD Research Centre | May 2020 - May 2021

Built domain-aware time-series forecasting and scenario analysis across energy, manufacturing, healthcare, finance, retail, telecommunications, automotive, and other industry intelligence centers.

Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on ~1 million images in ~3 hours, with 90% classification accuracy. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

Built an ML-powered lead-conversion prediction system and business-intelligence applications that moved sales prioritisation from intuition to data.

Achievements

15 honors and awards

Best Buy Talent Quotient (2026); Best Buy client AML Hall of Fame (2024), during the Wipro engagement.

Vehicle Routing & Transport Planning

Accenture | Google OR-Tools | Operations research

Designed vehicle-routing and mixed-integer models covering capacity, time windows, waiting time, demand, and departure constraints, then automated daily operational planning.

Industry Forecasting Platform

GD Research Centre | Time series | Scenario analysis

Designed and deployed domain-aware forecasting across 15+ industries, combining feature engineering, time-series modelling, and scenario analysis for enterprise decisions.

System Ownership

Design | Implementation | Production operations | Maintenance

Owned AI/ML systems from data and training pipelines through evaluation, deployment, monitoring, maintenance, incident support, and continuous improvement.

Engineering Leadership & Team Building

Architecture | Mentoring | Hiring | Cross-functional delivery

Stayed hands-on while leading ML/MLOps delivery, mentoring engineers, coordinating client and platform stakeholders, and helping scale the India AI engineering hub during the Wipro engagement through 300+ résumé screens and 50+ technical interviews.